

Joongmin Shin

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Research Interests

I am a Senior Researcher at the Human-Inspired AI Research Institute, Korea University, where I founded and lead **KUDoc**. Co-advised by **Prof. Jaehyung Seo** and **Prof. Heuseok Lim**, my research starts from real-world needs and asks how advances in AI capability can translate into systems that people can use effectively and rely on.

My training in parsing taught me to think structurally; my later work in human-inspired AI taught me what that structure should ultimately serve. I have since developed a connected research program spanning multimodal reasoning and agents, using structure, retrieval, knowledge representation, and memory as methodological tools for building AI systems that remain useful and reliable in practice. For my Ph.D., I aim to extend this work toward systems that reason over increasingly complex multimodal inputs and longer interactions.

My research has yielded **15 publications**, including **6 first-author papers** at NeurIPS, ACL, CVPR, and EMNLP, with main-track publications at all four venues and an ACL 2026 paper selected for **oral presentation**. I also hold **5 patents** and have received **7 awards**, including 3 Outstanding Paper or Research Awards.

Multimodal AI Parsing · Retrieval · Reasoning over documents and video

Agentic AI Knowledge Representation · Memory · Verification

Selected Publications

* First author or co-first author

ADMIT: Support-Gated Memory-Write Admission for Document QA Agents

Joongmin Shin*, Gyuho Shim, Hyeonseok Moon, Jaehyung Seo

NeurIPS 2026 – Conference on Neural Information Processing Systems (Main)

PILAR: A Page-Grounded Unified Evidence Representation via an Entity-Linked Assertion Graph for Open-Domain QA Agents over Multimodal Document Corpora

Joongmin Shin*, Gyuho Shim, Jung-Hun Lee, Jaehyung Seo

EMNLP 2026 – Conference on Empirical Methods in Natural Language Processing (Findings)

HiKEY: Hierarchical Multimodal Retrieval for Open-Domain Document Question Answering

Joongmin Shin*, Gyuho Shim, Jeongbae Park, Jaehyung Seo, Heuseok Lim

ACL 2026 – Annual Meeting of the Association for Computational Linguistics (Main, Oral)

M3DocDep: Multi-modal, Multi-page, Multi-document Dependency Chunking with Large Vision-Language Models

Joongmin Shin*, Jeongbae Park, Jaehyung Seo, Heuseok Lim

CVPR 2026 – IEEE/CVF Conference on Computer Vision and Pattern Recognition (Main)

MultiDocFusion: Hierarchical and Multimodal Chunking Pipeline for Enhanced RAG on Long Industrial Documents

Joongmin Shin*, Chanjun Park, Jeongbae Park, Jaehyung Seo, Heuseok Lim

EMNLP 2025 – Conference on Empirical Methods in Natural Language Processing (Main)

Intelligent Predictive Maintenance RAG Framework for Power Plants: Enhancing QA with StyleDFS and Domain Specific Instruction Tuning

Seongtae Hong*, Joongmin Shin*, Jaehyung Seo, Taemin Lee, Jeongbae Park, Cho Man Young, Byeongho Choi, Heuseok Lim

EMNLP 2024 – Conference on Empirical Methods in Natural Language Processing (Industry Track)

Search-Based Generation Techniques for Improving LLM Responses: A Comparative Study of Zero-shot and RAG on GPT-3.5 and GPT-4

Joongmin Shin*, SeongReol Park, Jung-Hun Lee

KIICE 2023 – Korea Institute of Information and Communication Engineering

Award: Outstanding Paper Award

EDT5: Proposed Embedding Model of T5 Encoder-Decoder Structure

Joongmin Shin*, Jugyung Jung, Jung-Hun Lee, Sanghyun Cho, Minho Kim, Miyeon Kim, Hyuk-Chul Kwon

KSC 2022 – Korea Software Congress

Award: Outstanding Paper Award

A Dependency Parsing Model with Reinforced Head-Dependent Constraint Rules: Combining Deep Learning and Linguistic Knowledge

Joongmin Shin*, Hyuk-Chul Kwon
HCLT 2022 – Conference on Human and Cognitive Language Technology
Award: Outstanding Research Award

Research Experience

KUDoc: Korea University Document AI Research Group Seoul, Republic of Korea
FOUNDER & LEAD 2024 – Present

- **Founding & research agenda:** Built the group from the ground up as an independent line of document AI work at Korea University.
- **Project coordination:** Oversaw KUDoc-affiliated projects that produced 15 publications and multiple industry collaborations.
- **Technical direction:** Steered research across multimodal document parsing, structure-aware retrieval, grounded document QA, and agentic systems.

Human-Inspired AI Research Institute, Korea University Seoul, Republic of Korea
SENIOR RESEARCHER Jun 2023 – Present

- **Structure-grounded multimodal reasoning:** Developed retrieval and reasoning methods that preserve document structure – layout, tables, figures, and cross-page dependencies – turning noisy multi-page documents and video into grounded evidence for QA.
- **First-author work:** Led first-author papers at NeurIPS, ACL, CVPR, and EMNLP on agent memory-write admission, page-grounded evidence graphs, hierarchical retrieval, multimodal document dependency parsing, and structure-preserving chunking.
- **Hybrid reasoning systems:** Built QA pipelines combining LLMs/LVLMs, knowledge graphs, symbolic rules, and evidence filtering for accurate, interpretable answers.
- **Applied validation:** Validated multimodal RAG in insurance, manufacturing, and healthcare, including an insurance document QA MVP with Samsung Fire & Marine.

AI Lab, Pusan National University Busan, Republic of Korea
GRADUATE RESEARCHER Sep 2020 – Feb 2023

- **Korean dependency parsing:** Developed neural-symbolic dependency parsing methods that integrate morphological and syntactic information for robust Korean language understanding.
- **Table and document QA:** Designed pipelines that convert semi-structured tables and documents into structured representations for question answering and interpretable reasoning.
- **Applied collaborations:** Contributed to Korean TTS prosody prediction with KT AI2XL and additional projects in finance QA, sentiment-based marketing, and recommendation systems.

Education

Pusan National University Busan, Republic of Korea
M.S. IN ARTIFICIAL INTELLIGENCE Mar 2021 – Feb 2023

- **Advisor:** Prof. Hyuk-Chul Kwon | AI Research Lab
- **Thesis:** A Neural-Symbolic Model for Overcoming Deep Learning Limitations in Korean Dependency Parsing
- **Selected outcomes:** Developed neural-symbolic Korean dependency parsing and table/document QA methods; received Outstanding Paper and Student-led Research Project awards.

Pusan National University Busan, Republic of Korea
B.S. IN COMPUTER SCIENCE AND ENGINEERING Mar 2016 – Feb 2021

- **Undergraduate Research:** Prof. Jeongu Kim’s group and Prof. Hyuk-Chul Kwon’s AI Research Lab
- **Selected outcomes:** Worked on knowledge-graph-based Korean NLP, Korean grammar correction, Korean TTS, and transformer-based Korean language models; received a merit-based Milestone Scholarship.

Selected Honors & Awards

2024	Deputy Prime Minister and Minister of Education Award , Competition for Best Practices in Academia–Industry Cooperation	National Research Foundation (NRF)
2023	Outstanding Paper Award , Korea Institute of Information and Communication Engineering (KIICE)	KIICE
2023	Grand Prize , Samsung Fire & Marine Insurance LLM-based MVP Model Idea Competition	Samsung
2023	Excellence Award (Technology Track) , KU-Grant Program	Korea University
2022	Outstanding Paper Award , Korea Software Congress (KSC)	KSC
2022	Outstanding Research Award , Student-led Research Project, Graduate School of AI	Pusan National University
2020	Milestone Scholarship (Grade A) , SW-centered University Milestone Scholarship (merit-based)	Pusan National University

Patents

2026	Hierarchical Multimodal Retrieval for Open-Domain Document Question Answering	Korea, Pending
2026	Multi-modal, Multi-page, Multi-document Dependency Chunking with Large Vision-Language Models	Korea, Pending
2025	Device and Method for Document Chunking, and Question-Answering Device and Method Using the Same	Korea · DP-2025-0093, Pending
Feb 2023	Rule Filtering Techniques and Methods for Knowledge Inference Systems Based on Deep Learning	Korea · P2022-0340-KR00, Registered
Nov 2022	AI-Based Adaptive Problem Recommendation System and Method for Learner-Level Assessment	Korea · 10-2022-0068075, Registered

Technical Skills

Modeling	LLM/LVLM fine-tuning, hierarchical retrieval, knowledge graphs, hybrid neural-symbolic reasoning, rule-based evidence filtering
Document AI	Multimodal parsing, hierarchical chunking, layout-aware processing, table/document QA
Programming	Python, SQL, C/C++, Java
ML & LLM Frameworks	PyTorch, HuggingFace Transformers, PEFT/LoRA, Accelerate, OpenAI/Anthropic API
Retrieval, RAG & Evaluation	FAISS, Elasticsearch, Pinecone, LangChain, LlamaIndex, lm-eval-harness
Document Processing	PyMuPDF, pdfplumber, Tesseract, PaddleOCR, LayoutLMv3, Donut
Languages	Korean (native), English (professional working proficiency)

Teaching & Mentoring

KUDoc, Korea University	Seoul, Republic of Korea
RESEARCH MENTOR	2024 – Present
- Mentor 3 M.S. students and 2 Ph.D. students on document AI, multimodal RAG, and structured evidence reasoning. - Co-authored 15 publications and 4 manuscripts currently under review at top-tier venues.	
Pusan National University	Busan, Republic of Korea
TEACHING ASSISTANT & RESEARCH MENTOR	2019 – 2022
Capstone research mentoring (2021–2022): Mentored undergraduate capstone teams on Korean dependency parsing and Table QA; one team achieved SOTA on the KLUE-DP benchmark. PNU Software Hackathon (2021): As TA for Prof. Hyuk-Chul Kwon’s Software Design course, mentored 100+ students across 30+ projects; seven student teams won awards including the Grand Prize. Course teaching assistant: Software Design (Android, Java) and Web Application Programming under Prof. Hyuk-Chul Kwon; Programming Principles (C) under Prof. Eunmi Jeong.	

Selected Research Projects

MULTIMODAL REASONING & DOCUMENT AI

Video Curation System – Multimodal Video QA

LEAD RESEARCHER · KOREA UNIVERSITY	Jul 2025 – Present
Video QA and RAG system for structured curation and grounded retrieval over long multimodal content.	

DocGraph Copilot – Structure-Aware Multimodal RAG

LEAD RESEARCHER · KOREA UNIVERSITY	Jan 2025 – Present
Structure-aware multimodal chunking pipeline for grounded retrieval and long-document QA; produced first-author papers at CVPR 2026 and EMNLP 2025.	

Science-Domain Korean LLM & RAG Pipeline

LEAD RESEARCHER · KOREA UNIVERSITY	Feb 2024 – Dec 2024
Korean LLM and RAG pipeline specialized for the science domain; led data preprocessing and chunking methodology development; produced an EMNLP 2024 Industry Track publication.	

Synerpeace Pilot – Insurance Document Assistant

LEAD RESEARCHER · KOREA UNIVERSITY · SAMSUNG FIRE & MARINE INSURANCE	Jul 2023 – Jan 2024
Insurance document QA pilot combining LLM-based retrieval with production-oriented application development.	

Synapse – Multilingual Medical Consultation System

LEAD RESEARCHER · KOREA UNIVERSITY · KU-GRANT PROGRAM

Jun 2023 – Present

Multilingual LLM-based medical consultation chatbot; selected for the KU-Grant research commercialization program.

FOUNDATION MODELS AND ADAPTATION

National Representative K-AI Research Team

RESEARCHER · KOREA UNIVERSITY · NC AI CONSORTIUM

Aug 2025 – Jan 2026

Multilingual foundation model development as a national-level representative researcher, spanning data pipelines, post-training, and evaluation.

Ko-Gemma – Korean-Adapted Gemma

RESEARCHER · KOREA UNIVERSITY

Feb 2024 – Mar 2024

Korean adaptation of Google’s Gemma model; contributed Korean-focused data adaptation and Korean benchmark evaluation.

KULLM – Korea University Korean LLM

RESEARCHER · KOREA UNIVERSITY

Dec 2023 – Feb 2024

Korean LLM initiative developed at Korea University; contributed instruction-following tuning and benchmark evaluation.

Robot Process Automation – PLC Control-Code Generation

RESEARCHER · KOREA UNIVERSITY · HYUNDAI MOBIS

Sep 2023 – Feb 2024

GPT-based assistant that generates PLC control code for smart-factory robot process automation, in collaboration with Hyundai Mobis.

Mi:deum K 1.0 – Korean Large Language Model

GRADUATE RESEARCHER · PUSAN NATIONAL UNIVERSITY · KT

Dec 2021 – Dec 2022

Contributed to KT’s Korean LLM ecosystem for enterprise generative AI; led model adaptation and Korean-targeted post-training evaluation.

Exobrain WiseQA – National QA

GRADUATE RESEARCHER · PUSAN NATIONAL UNIVERSITY · ETRI

Sep 2020 – Nov 2022

Table QA and neural-symbolic Korean sentence analysis for the national Exobrain project; introduced an Adapter-based architecture unifying separate text/table embedding models that reduced QA model parameter size by approximately 50%; research foundation for subsequent structure-grounded evidence work.

ADDITIONAL PROJECTS

- Korean parsing (96.28%) and TTS prosody (92.13%) systems with KT, deployed in KT GiGA Genie.
- Navigation-TTS for Navice in BMW/Mercedes-Benz.
- AI-based online judge with NextStage in PNU SW curriculum.
- AI mineral location prediction with KIGAM.

All Publications

* First author or co-first author

INTERNATIONAL TOP CONFERENCE PUBLICATIONS

ADMIT: Support-Gated Memory-Write Admission for Document QA Agents

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Intelligent Predictive Maintenance RAG Framework for Power Plants: Enhancing QA with StyleDFS and Domain Specific Instruction Tuning

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JOURNAL PUBLICATIONS

Multi-Paragraph Machine Reading Comprehension with Hybrid Reader over Tables and Text

Sanghyun Cho*, SeongReol Park, Hye-Lynn Kim, Jung-Hun Lee, Joongmin Shin, Hyuk-Chul Kwon
Applied Artificial Intelligence 38.1 (2024), p. 2367820. Taylor & Francis

Distance Based Korean WordNet (alias. KorLex) Embedding Model

SeongReol Park*, Joongmin Shin, Sanghyun Cho, Hyuk-Chul Kwon, Jung-Hun Lee
Applied Artificial Intelligence 38.1 (2024), p. 2398920. Taylor & Francis

DOMESTIC CONFERENCE PUBLICATIONS

Search-Based Generation Techniques for Improving LLM Responses: A Comparative Study of Zero-shot and RAG on GPT-3.5 and GPT-4

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KSC 2022 – Korea Software Congress
Award: Outstanding Paper Award

Evaluation of Generalization Performance in Korean Table Machine Reading Comprehension across Domain-Specific Datasets

Hyelin Kim*, Sanghyun Cho, Joongmin Shin, Hyuk-Chul Kwon
KSC 2022 – Korea Software Congress

A Dependency Parsing Model with Reinforced Head-Dependent Constraint Rules: Combining Deep Learning and Linguistic Knowledge

Joongmin Shin*, Hyuk-Chul Kwon
HCLT 2022 – Conference on Human and Cognitive Language Technology
Award: Outstanding Research Award

Rule-Augmented Neural Network-Based Dependency Parsing

Joongmin Shin*, Sanghyun Cho, Bongwoo Nam, Hyuk-Chul Kwon
KCC 2022 – Korea Computer Congress

Korean Machine Reading Comprehension Using Continual Learning

Joongmin Shin*, Sanghyun Cho, Hyuk-Chul Kwon
HCLT 2021 – Conference on Human and Cognitive Language Technology

MASTER'S THESIS

A Neural-Symbolic Model for Overcoming Deep Learning Limitations in Korean Dependency Parsing

Joongmin Shin*, Hyuk-Chul Kwon
Master's Thesis, Pusan National University, 2023

Selected Manuscripts Under Review

FIRST-AUTHOR MANUSCRIPTS

- **ICLR 2027.** Evaluating the practical utility of agentic memory.
- **ICLR 2027.** Citation grounding for video question answering.
- **TACL.** Evidence auditing and support-sensitive evaluation for multimodal QA.
- **IEEE TPAMI.** Error-propagation diagnostics for PDF-to-RAG pipelines.